DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

Test Booklet Series

A

TEST BOOKLET

GEOPHYSICIST

PAPER – II

T. B. C. : GPGM – 4/20

Sl. No. 40021

Time Allowed : 3 Hours

Maximum Marks : 300

: INSTRUCTIONS TO CANDIDATES :

1. IMMEDIATELY AFTER COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET **DOES NOT** HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET OF SAME SERIES ISSUED TO YOU.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES **A, B, C** OR **D**, AS THE CASE MAY BE, IN THE APPROPRIATE PLACE IN THE ANSWER SHEET USING BALL POINT PEN (BLUE OR BLACK).
3. You have to enter your **Roll No.** on the Test Booklet in the Box provided alongside. **DO NOT** write *anything else* on the Test Booklet.
4. **YOU ARE REQUIRED TO FILL UP & DARKEN ROLL NO., TEST BOOKLET / QUESTION BOOKLET SERIES IN THE ANSWER SHEET AS WELL AS FILL UP TEST BOOKLET / QUESTION BOOKLET SERIES AND SERIAL NO. AND ANSWER SHEET SERIAL NO. IN THE ATTENDANCE SHEET CAREFULLY. WRONGLY FILLED UP ANSWER SHEETS ARE LIABLE FOR REJECTION AT THE RISK OF THE CANDIDATE.**
5. This Test Booklet contains **150** items (questions). Each item (question) comprises four responses (answers). You have to select the correct response (answer) which you want to mark (darken) on the Answer Sheet. In case, you feel that there is more than one correct response (answer), you should mark (darken) the response (answer) which you consider the best. In any case, choose **ONLY ONE** response (answer) for each item (question).
6. You have to mark (darken) all your responses (answers) **ONLY** on the **separate Answer Sheet** provided by **using BALL POINT PEN (BLUE OR BLACK)**. See instructions in the Answer Sheet.
7. All items (questions) carry equal marks. All items (questions) are compulsory. Your total marks will depend only on the number of correct responses (answers) marked by you in the Answer Sheet. There will be negetive markings for wrong answers. **25 (Twenty Five) percent of marks alloted to a particular question will be deducted as negative marking for every wrong answer.**
8. Before you proceed to mark (darken) in the Answer Sheet the responses (answers) to various items (questions) in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per the instructions sent to you with your **Admission Certificate**.
9. After you have completed filling in all your responses (answers) on the Answer Sheet and after conclusion of the examination, you should hand over to the Invigilator the *Answer Sheet* issued to you. You are allowed to take with you the candidate's copy / second page of the Answer Sheet along with the **Test Booklet**, after completion of the examination, for your reference.
10. Sheets for rough work are appended in the Test Booklet at the end.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

SEAL

1. The D" layer in the interior of the Earth is found at the boundary between which of the two layers given below ?

(A) Crust and Mantle

(B) Upper and Lower Crust

(C) Mantle and Outer Core

(D) Inner and Outer Core

2. Which one of the following hotspots is located nearest to the junction of the South American, African and the Antarctic Plates ?

(A) Kerguelen

(B) Bouvet

(C) Reunion

(D) Conrad

3. Which one of the given co-ordinates is associated with the Earth's geomagnetic north pole ?

(A) $79^{\circ}N, 71^{\circ}W$

(B) $79^{\circ}S, 109^{\circ}E$

(C) $76^{\circ}N, 101^{\circ}W$

(D) $66^{\circ}S, 141^{\circ}E$

4. Magnetic measurements were made on a basalt flow at present at $55^{\circ}N$, $80^{\circ}E$. The angle of inclination of the remanent magnetization of this basalt is 60°. The magnetic latitude of this site at the time the basalt was magnetized was :

(A) $21^{\circ}N$

(B) $31^{\circ}N$

(C) $41^{\circ}N$

(D) $51^{\circ}N$

5. In the Magnetic Polarity Time Scale, the present-day normal Epoch / Chron is known as :

(A) Gilbert

(B) Gauss

(C) Matuyama

(D) Brunhes

6. The most recent magnetic reversal occurred around :

(A) 0.78 Ma

(B) 0.92 Ma

(C) 1.01 Ma

(D) 1.78 Ma

7. Seismic S-waves are characterised by :

(A) Velocities slower than seismic P-waves

(B) No volume change

(C) Lack of ability to sustain shear in liquids

(D) All of these

8. The Nafe-Drake Curve defines the relationship between seismic velocities and :

(A) Stress

(B) Strain

(C) Density

(D) Ground acceleration

9. Ground roll, similar to how ocean waves propagate, during earthquakes is caused by :

(A) P-waves

(B) Rayleigh waves

(C) S-waves

(D) Love waves

10. If d is the distance from the source of an earthquake, the amplitude of a seismic body wave is proportional to :

(A) d^2

(B) $\sqrt{d}$

(C) $\frac{1}{d}$

(D) d

11. In general, the correct first-to-last sequence of arrival of seismic waves on a seismogram is :

(A) P-waves, S-waves, Rayleigh waves, Love waves

(B) P-waves, Love waves, S-waves, Rayleigh waves

(C) P-waves, Love waves, Rayleigh waves, S-waves

(D) P-waves, S-waves, Love waves, Rayleigh waves

12. Great earthquakes have a magnitude :

(A) > 8.0

(B) 7.0 – 7.9

(C) 6.0 – 6.9

(D) 5.0 – 5.9

13. The presence of PKJKP waves in the Earth's interior is the direct evidence that the Earth's :

(A) Outer core is molten

(B) Inner core is solid

(C) Outer core is solid

(D) Innter core is molten

14. The discontinuity between the Earth's Inner Core and Outer Core is known as the :

(A) Lehman discontinuity

(B) Guttenberg discontinuity

(C) Repetti discontinuity

(D) Conrad discontinuity

15. The Indian Ocean region near the southern tip of mainland India and Sri Lanka is characterized by :

(A) Positive geoid anomaly

(B) Negative geoid anomaly

(C) Zero geoid anomaly

(D) None of these

16. For seabed topography, the difference between the magnitudes of the geoid height anomalies calculated for the Pratt compensation model and the Airy model is typically :

(A) Positive

(B) Negative

(C) Zero

(D) None of these

17. The oldest patch of undisturbed oceanic crust on Earth lies in the Mediterranean Sea with an age (in million years) of about :

(A) 80

(B) 160

(C) 190

(D) 340

18. Maximum continental crust thickness is generated at :

(A) Divergent plate boundaries

(B) Convergent plate boundaries

(C) Sinistral strike-slip plate boundaries

(D) Dextral strike-slip plate boundaries

19. Within the Earth, heat moves through the lithosphere (both oceanic and continental) predominatly by :

(A) Conduction

(B) Advection

(C) Convection

(D) Radiation

20. Most of the total heat loss from the Earth occurs due to :

(A) Volcanic eruptions

(B) Creation and cooling of oceanic crust

(C) Melting of glaciers

(D) Anthropogenic activities

21. Which one of the following is the main internal heat source for the Earth as a whole ?

(A) Solar Radiation

(B) Frictional heating

(C) Radioactive heat

(D) Meteorite impact

22. Milne's seismograph was useful only for recording :

(A) Great earthquakes

(B) Strong earthquakes

(C) Moderate earthquakes

(D) Minor earthquakes

23. Seismic Risk is defined as :

(A) Hazard / Vulnerability

(B) Vulnerability / Hazard

(C) Hazard × Vulnerability

(D) None of these

24. The evolution of foreshocks and aftershocks in relation with an earthquake main shock is described by :

(A) Byerlee's Law

(B) Omori's Law

(C) Coulomb's Law

(D) Amonton's Law

25. The determination of the level of severity associated with strong ground motion during earthquakes requires the evaluation of which seismic wave properties listed below ?

(A) Frequency content

(B) Amplitude

(C) Effective duration

(D) All of these

26. Liquefaction in a granular soil during an earthquake occurs because of :

(A) Decrease in pore water pressure

(B) Increase in effective stress

(C) Increase in pore water pressure

(D) None of these

27. In structures not designed for earthquake resistance, damage induced by moderate earthquakes starts to occur typically when ground accelerations reach around :

(A) 1 cm/sec^2

(B) 10 cm/sec^2

(C) 100 cm/sec^2

(D) 1000 cm/sec^2

28. In the revised earthquake zoning map of India that divides India into 4 seismic zones (zone 2, 3, 4 and 5), zone 5 has been assigned a zone factor of :

(A) 0.46

(B) 0.36

(C) 0.24

(D) 0.16

29. The instrument used for making measurements in strong-motion seismology is :

(A) Oscillograph

(B) Seismograph

(C) Telegraph

(D) Accelerograph

30. Given the same earthquake magnitude, distance to the site, and site conditions, which one of the statement about the ground motion produced by thrust, strike-slip and normal earthquakes is false ?

(A) Ground motions from strike-slip earthquakes tend to be larger than those from thrust earthquakes

(B) Ground motions from normal faulting earthquakes tend to be smaller than those from strike-slip earthquakes

(C) Ground motions from thrust earthquakes tend to be larger than those from normal faulting earthquakes

(D) None of these

31. Which one of the following statements is false for an electric charge in classical Electrodynamics ?

(A) It is a scalar quantity

(B) It is measurable

(C) It can exist without mass

(D) It can be positive or negative

32. In classical Electrodynamics, Maxwell's equation are a system of coupled :

(A) Second order, linear partial differential equations

(B) First order, linear partial differential equations

(C) Second order, non-linear partial differential equations

(D) First order, non-linear partial differential equations

33. The short wavelength component of the non-dipole magnetic field at the Earth's surface originates dominantly in the :

(A) Crust

(B) Mantle

(C) Outer Core

(D) Inner Core

34. As the latitude increases, the angle at which the Earth's magnetic field lines cross the Earth's surface :

(A) Remains constant

(B) Decreases

(C) Increases

(D) Varies randomly

35. The geomagnetic field has an average intensity (in nanotesla) of around :

(A) 1000

(B) 5000

(C) 10,000

(D) 50,000

36. Maps of the geographic distribution of the non-dipole field (NDF) show a large anomalous region, where the NDF has a strong negative value, over the :

(A) Indian Ocean near Sri Lanka

(B) Atlantic Ocean east of South America

(C) Pacific Ocean west of North America

(D) Arctic Ocean

37. The region in which a planet's magnetic field is stronger than the interplanetary field is called the :

(A) Magnetopause

(B) Magnetosheath

(C) Magnetosphere

(D) Magnetotail

38. Telluric currents in the conducting layers of the Earth :

(A) Are induced by Sq ionospheric fields

(B) Cause secondary magnetic fields

(C) Give us information on the electrical conductivity of the lithosphere

(D) All of these

39. The geometry of geomagnetic field lines is the most complex inside the :

(A) Core

(B) Mantle

(C) Crust

(D) Lithosphere

40. Spherical harmonic analysis is used in modern geophysics to describe the global distributions of geophysical properties such as :

(A) Magnetism

(B) Gravity

(C) Thermal

(D) All of these

41. Which one of the following statement is true for the blocking temperature Tb and Curie temperature Tc of magnetic minerals ?

(A) $Tb = Tc$

(B) $Tb < Tc$

(C) $Tb > Tc$

(D) $Tb \geq Tc$

42. In general, as the temperature rises, resistivity in rocks is seen to :

(A) Increase

(B) Decrease

(C) Remain constant

(D) Vary randomly

43. Archie's Law states that in water bearing, porous sediments with insignificant clay content, the formation resistivity is directly proportional to the :

(A) Sediment porosity

(B) Water saturation

(C) Resistivity of the water in the pores

(D) Sediment porosity and water saturation

44. In resistivity surveys on uniform ground, the fraction of the current that typically penetrates below a depth equal to the separation of the current electrodes is about :

(A) 70%

(B) 50%

(C) 30%

(D) 10%

45. Vertical Electric Sounding in a layered rock section involves refraction of current paths. The incident path $\mu 1$ (at an angle $\theta 1$ with the normal to the layering) and the refracted path $\mu 2$ (at an angle $\theta 2$ with the normal to the layering) at an interface separating rocks of different resistivities are related by the equation :

(A) $\mu 1 \tan \theta 1 = \mu 2 \tan \theta 2$

(B) $\mu 1 \tan \theta 2 = \mu 2 \tan \theta 1$

(C) $\mu 1 \sin \theta 1 = \mu 2 \sin \theta 2$

(D) $\mu 1 \sin \theta 2 = \mu 2 \sin \theta 1$

46. Equal spacing between the potential (P) and current (C) electrodes during resistivity survey is used in the :

(A) Dipole-Dipole array

(B) Wenner array

(C) Gradient array

(D) Schlumberger array

47. Four electrodes are used in electrical survey to avoid contact _____ of the electrodes.

(A) Resistance

(B) Potential

(C) Current

(D) Friction

48. Electrical imaging is sometimes done by pseudosection surveying. A pseudosection is so called because the :

(A) Orientation of the section is arbitrary

(B) Depth scale of the section is arbitrary

(C) Horizontal distance scale of the section is arbitrary

(D) Resistivity meter measures apparent resistivity

49. Vertical Electric Sounding is most appropriate in the resistivity survey of a layered section containing :

(A) Synformally folded rocks with a vertical axial plane

(B) Antiformally folded rocks with a vertical axial plane

(C) Uniformly steeply dipping rocks

(D) Undeformed horizontal section of rocks

50. Which one of the following has the lowest resistivity when dry ?

(A) Granite

(B) Sand

(C) Clay

(D) Gabbro

51. In which of the following surveys are dangerous voltages and currents used that may be potentially lethal ?

(A) Induced polarization

(B) Self-potential

(C) Spontaneous polarization

(D) Resistivity

52. Induced polarization survey cannot be used to detect which one of the following sulphide ores ?

(A) Zn

(B) Ni

(C) Fe

(D) Co

53. The types of electrode combinations used in Self-potential surveys is : (C – Current electrode ; P – Potential electrode)

(A) 2C and 2P

(B) 2C

(C) 2P

(D) 1C and 1P

54. Select from the list the type of electrode(s) used in induced polarization surveys :

(C – Current electrode ; P – Potential electrode)

(A) Metal

(B) Nonpolarizing

(C) Metal P electrodes, nonpolarizing C electrodes

(D) Metal C electrodes, nonpolarizing P electrodes

55. The maximum depth below which Self-potential anomalies are unlikely to detect any ore body is typically about :

(A) 5 m

(B) 10 m

(C) 20 m

(D) 30 m

56. If the fresh water in the pores of a disseminated lead sulphide ore were replaced by saline water, how would the values of the frequency effect (FE) and metal factor (MF) in an Induced polarization survey be affected ?

(A) Both increase

(B) FE increases MF remains constant

(C) FE remains constant MF increases

(D) Both decrease

57. The Turam electromagnetic survey system is a :

(A) Moving-source, moving-receiver system

(B) Moving-source, fixed-receiver system

(C) Fixed-source, moving-receiver system

(D) Fixed-source, fixed-receiver system

58. In massive sulphide deposits that generate spontaneous polarization, a typical value of the measured Self-potential is in the order of a few :

(A) Millivolts

(B) 10s of millivolts

(C) 100s of millivolts

(D) Volts

59. Which one of the following statement is false about electromagnetic survey ?

(A) Operates when the surface layer has a high resistivity

(B) Gives more precise location of ore bodies than resistivity survey

(C) Easier to use than resistivity survey as electrodes do not have to be moved

(D) Less penetration when the surface layer is very conductive

60. Which one of the following statements is false about the Slingram electromagnetic survey system ?

(A) Transmitter and receiver coils are connected to the ground

(B) Transmitter and receiver are connected by a cable

(C) Transmitter and receiver separation is kept constant during a traverse

(D) Singram system can be used for airborne survey

61. Which one of the following statement is true about the Transient electromagnetic method (TEM) ?

(A) The transmitter signal needs to be accurately cancelled at the receiver

(B) The receiver signal is measured after the transmitter has been switched off

(C) The TEM method cannot be used for aerial surveys

(D) The receiver signal goes instantaneously to zero when the transmitter is switched off if no conducting target exists

62. Skin depth is defined as the distance an e-m wave must travel before its amplitude decays by a factor of : (e is exponential)

(A) 1/2

(B) 1/2e

(C) 1/e

(D) $1/e^2$

63. Radio and cellphones do not work in a tunnel mainly because e-m waves are :

(A) Unable to enter the tunnel opening

(B) Reflected multiple times within the tunnel

(C) Damped by the noise in the tunnel

(D) Absorbed as they travel through the rocks

64. For surveying deepest targets using e-m waves, the best solution is to use :

(A) Ultra high frequency

(B) High frequency

(C) Very low frequency

(D) Any of these

65. The Very-Low-Frequency (VLF) surveying method is best suited for mapping contacts or sheet like bodies that dip around :

(A) 0°

(B) 30°

(C) 60°

(D) 90°

66. The Very-Low-Frequency (VLF) receiver detects the combined fields of transmitter and target in which the transmitter field is :

(A) Horizontal

(B) Tilted

(C) Vertical

(D) Sinusoidal

67. During a Ground-Penetrating Rader (GPR) survey, the first wave to arrive at the receiver is the :

(A) Direct ground wave

(B) Direct air wave

(C) Critically refracted wave

(D) Reflected wave

68. The velocity of the radar waves in Ground-Penetrating Radar (GPR) survey is proportional to :

(ε_r – Relative permittivity)

(A) ε_r

(B) $\sqrt{\varepsilon_r}$

(C) $\frac{1}{\varepsilon_r}$

(D) $\frac{1}{\sqrt{\varepsilon_r}}$

69. Ground-Penetrating Radar (GPR) penetration will be least in which one of the following materials ?

(A) Shale

(B) Dry Sand

(C) Sea Water

(D) Pure Limestone

70. Ground-Penetrating Radar (GPR) survey offers sections that are of :

(A) Hight resolution but low depth penetration

(B) Low resolution and low depth penetration

(C) High resolution and high depth penetration

(D) Low resolution but high depth penetration

71. A profile across a known subsurface body fails to show an anomaly. This may be because the :

(A) Station spacing was too small

(B) Signal-to-noise ratio was too large

(C) Instrument was very sensitive

(D) Data had not been reduced

72. An area has been surveyed along traverses and some of the traverses intersect. The most appropriate way to display the results would be by :

(A) Contouring

(B) Stacked profiles

(C) Fence projection

(D) Isometric projection

73. Stacking helps to :

(A) Display the results from measurements made along widely and irregularly spaced traverses

(B) Improve the signal-to-noise ratio

(C) Reduce the data

(D) All of these

74. In geophysical techniques a body near the surface, when compared to the same body buried deeper, gives a :

(A) Narrower anomaly

(B) Broader anomaly

(C) Weaker anomaly

(D) Negative anomaly

75. The brightness of daylight at a place is recorded continuously for about a week and then Fourier analyzed. What is the frequency (in Hz) of the lowest period found ?

(A) 0.04167000

(B) 0.00069440

(C) 0.00001157

(D) None of these

76. When sampling a harmonic at an interval of 0.02 second, what Nyquist frequency must be used to prevent aliasing ?

(A) 100 Hz

(B) 75 Hz

(C) 50 Hz

(D) 25 Hz

77. When a geophysical signal is sampled at intervals, the apparent wavelength is always longer than the true wavelength of the signal once the sampling interval exceeds :

(A) 1/4 the wavelength

(B) 1/2 the wavelength

(C) 3/4 the wavelength

(D) The wavelength

78. Which filter attenuates any frequency outside the defined pass band ?

(A) Band-pass filter

(B) Band-reject filter

(C) Band-stop filter

(D) All of these

79. The Navier-Stokes equation describes the motion of :

(A) Solid substances

(B) Non-viscous fluids

(C) Viscous fluids

(D) None of these

80. The Navier-Stokes equation can be used to model the :

(A) Weather

(B) Ocean Currents

(C) Air flow around a wing

(D) All of these

81. Which one of the following is not a numerical technique to solve partial differential equations ?

(A) Finite Element method

(B) Runge-Kutta method

(C) Finite Difference method

(D) Boundary Element method

82. Discretization in the Finite Element method involves the division of a body into a mesh consisting of numbered :

(A) Elements

(B) Nodes

(C) Nodes and Elements

(D) Nodes or Elements

83. Shape functions used in Finite Element method are :

(A) Iterative functions

(B) Interpolation functions

(C) Mesh area functions

(D) Matrix function

84. Which one of the following is false for the Boundary Element Method ?

(A) Involves volume discretization in 3D into elements

(B) Adept at investigations of crustal discontinuities

(C) Heterogeneous material properties modelled by contact boundary

(D) Based on the solution to Kelvin's problem for point load related deformation in a body

85. The maximum radiation of the sun occurs around the wavelength :

(A) 0.15 μm

(B) 0.35 μm

(C) 0.55 μm

(D) 0.75 μm

86. Remote Sensing of the Earth's surface occurs through atmospheric windows that include the wavelength region :

(A) 1.35 – 1.45 μm

(B) 2.7 – 2.9 μm

(C) 3.7 – 4.1 μm

(D) 7.0 – 15.0 μm

87. Which one of the following Bands of Synthetic Aperture Radar (SAR) has the largest penetrative depth ?

(A) C band

(B) L band

(C) X band

(D) S band

88. The attenuation length of an Electromagnetic radiation is defined as the length over which the power of the radiation falls by the combined effect of absorption and scattering by a factor of : (e is exponential)

(A) e

(B) e^2

(C) 1/e

(D) $1/e^2$

89. Earth is a soiurce for passive remote sensing because it emits electromagnetic radiation as a black body at 300°K with a peak wavelength at around :

(A) 0.15 μm

(B) 1.35 μm

(C) 4.50 μm

(D) 9.50 μm

90. Assuming that the Earth is a black body at 300°K, the emitted radiation of the Earth exceeds the solar radiance beyond a wavelength of about :

(A) 1 μm

(B) 3 μm

(C) 5 μm

(D) 7 μm

91. Which one of the following is not used in Remote Sensing ?

(A) X-Rays

(B) Infrared

(C) Microwave

(D) Radiowave

92. Lambertian reflectors or emitters obey which one of the laws given below. $I(\theta)$ is the intensity of the reflected or the emitted electromagnetic radiation as a function of the angle from the perpendicular θ, and I_0 is the intensity at $\theta = 0°$.

(A) $I(\theta) = I_0 \sin \theta$

(B) $I(\theta) = I_0 \cos \theta$

(C) $I(\theta) = I_0 \cot \theta$

(D) $I(\theta) = I_0 \tan \theta$

93. Which one of the following is a Radiometric quantity in remote sensing ?

(A) Radiance

(B) Irradiance

(C) Exitance

(D) All of these

94. Which one of the statement about the Bidirectional Reflectance Distribution Function (BRDF) is false ?

(A) It is dimensionless

(B) It describes the reflectance of a surface

(C) It is a function of view geometry

(D) It is a function of illumination geometry

95. Which one of the following does not occur in the visible spectrum region when sunlight falls on a leaf ?

(A) Absorption in the ~ 0.45 μm region

(B) Reflectance in the ~ 0.55 μm region

(C) Absorption in the ~ 0.67 μm region

(D) Absorption in the ~ 1.4 μm region

96. Water bodies rich in phytoplankton appear green in color because phytoplankton :

(A) Contain chlorophyll that absorbs strongly near 0.443 μm region

(B) Contain chlorophyll that absorbs weakly in the 0.52 and 0.55μm regions

(C) Content increase causes increase in backscattering in the green region

(D) All of these

97. Which one of the following is an example of an active remote sensor operation in the microwave region ?

(A) Scanning Microwave Radiometer (SMR)

(B) Synthetic Aperature Radar (SAR)

(C) Liner Imaging Self-scanning Sensor (LISS)

(D) Laser Imaging Detection and Ranging Sensor (LIDAR)

98. In optical imaging systems, the Airy disc and the Airy pattern resulting from an uniformly illuminated, circular aperture is produced due to :

(A) Refraction

(B) Reflection

(C) Diffraction

(D) Absorption

99. Two objects that are point sources of light can be resolved spatially using the Rayleigh criterion which states that the objects can just be resolved when the peak of the Airy disc for the first object occurs at the :

(A) First minima of the Airy disc of the second object

(B) First maxima of the Airy disc of the second object

(C) Second minima of the Airy disc of the second object

(D) Second maxima of the Airy disc of the second object

100. Spectral resolution of a sensor refers to which of the following statement ?

(A) It is related to the bandwidth over which the sensor gathers information

(B) Determines the smallest features that can be resolved in a scene

(C) It is inversely proportional to bandwidth

(D) All of these

101. Which one of the following Thematic Mapper Spectral Bands is used for Surface temperature mapping ?

(A) 2.08 – 2.35 μm

(B) 10.4 – 12.5 μm

(C) 1.55 – 1.75 μm

(D) 0.76 – 0.90 μm

102. In Geographical Information System (GIS) attribute data is :

(A) Raster data

(B) Non-spatial data

(C) Location data

(D) Vector data

103. For a high spatial resolution, the Instantaneous Field of View (IFOV) is reduced. This leads to a poor :

(A) Temporal Resolution

(B) Spectral Resolution

(C) Radiometic Resolution

(D) All of these

104. For constant Instantaneous Field of View (IFOV), the spatial resolution of an imaging instrument is :

(A) Directly proportional to its height from the target

(B) Inversely proportional to its height from the target

(C) Independent of its height from the target

(D) Going to remain constant

105. Geosynchronous satellites are placed in an orbit around the Earth at an altitude of about :

(A) 24,000 km

(B) 28,000 km

(C) 32,000 km

(D) 36,000 km

106. The Global Positioning System (GPS) uses Microwave signals in the :

(A) P band

(B) L band

(C) S band

(D) C band

107. The Global Positioning System (GPS) satellites are located at an altitude of about :

(A) 10,200 km

(B) 15,200 km

(C) 20,200 km

(D) 25,200 km

108. Signal propagation biases are induced in the Global Positioning System (GPS) due to presence of water vapor in the Earth's :

(A) Troposphere

(B) Ionosphere

(C) Mesosphere

(D) Thermosphere

109. The Global Positioning System (GPS) measures heights above the reference :

(A) Everest spheroid

(B) Mean Sea Level

(C) Geoid

(D) WGS 84 ellipsoid

110. For a raster data model in the Geographical Information System (GIS), which of the following statement is false ?

(A) Cells are the basic units for recording information.

(B) Data model is a spaghetti model.

(C) Data is stored in blocks of cells.

(D) Cell size determines ability to specify a location.

111. In a Diasaster Management Cycle, the actual disaster event occurs between which two stages ?

(A) Response and Recovery

(B) Recovery and Mitigation

(C) Preparedness and Response

(D) Mitigation and Preparedness

112. The Geographical Information System (GIS) is relevant to which stages of the Disaster Management Cycle ?

(A) Preparedness

(B) Recovery

(C) Mitigation

(D) All of these

113. In a Disaster Management Terminology, Hurricane Katrina in the US in 2005 and Super Typhoon Haiyan in the Philippines are examples of :

(A) Catastrophe

(B) Disaster

(C) Crisis

(D) Emergency

114. The Indian National Disaster Management Act (NDMA) was enacted by the Government in :

(A) 2002

(B) 2005

(C) 2007

(D) 2012

115. The Chairperson of the State Disaster Management Authorities (SDMAs) is the :

(A) Chief Minister

(B) Governor

(C) Home Minister

(D) Prime Minister

116. The Nodal Ministry for Disaster Management at National Level for Natural Disasters, except Drought, Epidemics and Civil Strife, is the :

(A) Ministry of Environment and Forest

(B) Ministry of Earth Sciences

(C) Ministry of Home Affairs

(D) Ministry of Mines

117. The highest magnitude instrumentally recorded earthquake is the :

(A) Tohoku Earthquake, Japan, 2011

(B) Great Alaskan Earthquake, 1964

(C) Kamchatka Earthquake, Russia, 1952

(D) Valdivia Earthquake, Chile, 1960

118. The probability of experiencing a certain intensity of natural or anthropogenic perilous event at a certain location is known in Disaster Management as :

(A) Vulnerability

(B) Hazard

(C) Exposure

(D) Risk

119. Maximum fatalities were recorded in which one of the following earthquakes ?

(A) Tangshan Earthquake, China, 1976

(B) Haiyuan Earthquake, 1920

(C) Shaanxi Earthquake, China, 1556

(D) Sumatra Earthquake, Indian Ocean, 2004

120. Maximum fatalities were recorded in which one of the following Tsunamis ?

(A) Indian Ocean Tsunami, 2004

(B) Tohoku Tsunami, Japan, 2011

(C) Messina Tsunami, Italy, 1908

(D) Lisbon Tsunami, Portugal, 1755

121. The Spanish Flu caused by the Influenza A/H1N1 viruses between 1918-1920 resulted in a worldwide death toll of at least :

(A) 5 million

(B) 12 million

(C) 17 million

(D) 200 million

122. The Covid-19 pandemic has resulted in a confirmed worldwide death toll (till June 2021) of about :

(A) 2 million

(B) 4 million

(C) 6 million

(D) 8 million

123. Probably the deadliest Tornado in recorded world history, that killed about 1300 people, occurred in 1989 in :

(A) United States

(B) Italy

(C) India

(D) Bangladesh

124. The deadliest tropical cyclone / typhoon ever recorded, with greater than 500000 fatalities, was :

(A) Bhola, 1970

(B) Nina, 1975

(C) Coringa, 1839

(D) Nargis, 2008

125. The most recent Super cyclonic storm in the North Indian Ocean was :

(A) Yaas

(B) Amphan

(C) Tauktae

(D) Kyarr

126. The Beaufort scale is a scale that measures :

(A) Aircraft height

(B) Aircraft speed

(C) Strength of winds

(D) Direction of winds

127. The Beaufort scale consists of Beaufort numbers that range from :

(A) 0-12

(B) 1-12

(C) 1-10

(D) 0-10

128. Which among the following is the most cyclone prone state in India between 1891-2000 ?

(A) Gujarat

(B) Maharashtra

(C) Tamil Nadu

(D) Odisha

129. Deep focus Earthquakes have a focal depth of :

(A) > 75 km

(B) Between 75 and 300 km

(C) > 300 km

(D) > 720 km

130. Volcanic hazard is not associated with which one of the plate boundary setting listed below ?

(A) Continental-oceanic convergent

(B) Continental-continental convergent

(C) Ocean-ocean convergent

(D) Ocean-ocean divergent

131. Volcanoes can be linked with other natural hazards such as :

(A) Landslides

(B) Tsunamis

(C) Fire

(D) All of these

132. Which one of the following is an active volcano in India ?

(A) Narcondam Island

(B) Dhinodhar Hills

(C) Barren Island

(D) Dhosi Hill

133. The bearing of a lineation is at 90° to the strike of the plane 132°, 55° SW containing it. The plunge amount of the lineation will be :

(A) 55°

(B) 35°

(C) 90°

(D) 48°

134. If the rake of a line on a plane is less than 90° then which one of the following relationships between the dip of the plane (φ) and the plunge of the line (θ) is true ?

(A) $\varphi = \theta$

(B) $\varphi > \theta$

(C) $\varphi < \theta$

(D) $\varphi \leq \theta$

135. On flat ground, the true thickness (t) of a bed is related to its width of outcrop (w) and dip (θ) through the relationship :

(A) $t = w \cos\theta$

(B) $t = w \tan\theta$

(C) $t = w \cot\theta$

(D) $t = w \sin\theta$

136. Rotation of a plane along its strike changes the plane's :

(A) Strike amount

(B) Strike direction

(C) Dip amount

(D) All of these

137. A syncline can also be a / an :

(A) Antiform

(B) Synform

(C) Tight fold

(D) All of these

138. The number of hinges present in a monocline is :

(A) 0

(B) 1

(C) 2

(D) 3

139. Porphyroclasts form due to grain-size reduction in which one of the following ?

(A) Mylonites

(B) Cataclasite

(C) Breccia

(D) Gouge

140. Cleavage is a type of :

(A) Bedding

(B) Foliation

(C) Schistosity

(D) Gneissosity

141. The kinematic vorticity number during deformation was measured to be 0.5. The deformation is interpreted as :

(A) Pure shear

(B) Simple shear

(C) Sub-simple shear

(D) Super-simple shear

142. The difference between mean stress and the total stress on a body is called the :

(A) Hydrostatic stress

(B) Differential stress

(C) Lithostatic stress

(D) Deviatoric stress

143. Typical natural geological strain rates are around :

(A) $10^{-17} - 10^{-18}\ s^{-1}$

(B) $10^{-14} - 10^{-15}\ s^{-1}$

(C) $10^{-11} - 10^{-12}\ s^{-1}$

(D) $10^{-7} - 10^{-8}\ s^{-1}$

144. A material with elastic-plastic rheology is known as a :

(A) Prandtl material

(B) Bingham material

(C) St. Venant material

(D) Maxwell material

145. Strength of the Lower Crust (~ 20 km – Moho) dominantly depends on the brittle-plastic transition of :

(A) Quartz

(B) Calcite

(C) Feldspar

(D) Olivine

146. Heave of a fault is defined as the horizontal component of the :

(A) Net slip vector

(B) Throw

(C) Dip slip vector

(D) Dip separation vector

147. When away from subduction zones, the seismogenic zone in the continental crust exists at depths typically between :

(A) 0-14 km

(B) 2-14 km

(C) 0-35 km

(D) 2-35 km

148. During an earthquake slip event the shape of the typical displacement profile along the seismogenic fault is :

(A) Bell-shaped

(B) Rectangular

(C) Triangular

(D) Irregular

149. According to the Anderson theory of faulting and conditions for plane strain, the two conjugate shear fractures intersect along which stress axes ?

(A) σ_1

(B) σ_2

(C) σ_3

(D) Any of these

150. According to the Anderson theory of faulting and conditions for plane strain, the vertical stress axis during thrust faulting is :

(A) σ_1

(B) σ_2

(C) σ_3

(D) Any of these

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK


SEAL